

1. Administrative & Study Identification

Full Study Title: An Open Label, Multi-center Phase IIb Study of Asciminib (ABL001) Monotherapy in Previously Treated Patients With Chronic Myeloid Leukemia in Chronic Phase (CML-CP) With and Without T315I Mutation **Short Title/Acronym:** AIM4CML Study
Protocol Number: CABL001AUS04 **NCT Number:** NCT04666259 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Asciminib (ABL001) / Scemblix® **Phase of Development:** Phase IIb **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety profile of asciminib 40 mg BID and 80 mg QD in patients with CML-CP without T315I mutation after ≥ 2 prior TKIs, and of asciminib 200 mg BID in patients with T315I mutation after ≥ 1 prior TKI over 24 weeks. **Secondary Objectives:** To evaluate the efficacy (Major Molecular Response [MMR] at 12, 24, 48, and 72 weeks), rate of MR4 and MR4.5, time to MMR, progression-free survival (PFS), overall survival (OS), and time to treatment failure.

2.2 Background & Rationale Chronic myeloid leukemia (CML) is driven by the BCR-ABL1 tyrosine kinase. Despite the success of ATP-competitive tyrosine kinase inhibitors (TKIs), many patients experience resistance or intolerance to successive lines of therapy, leaving them with limited treatment options (especially those with the T315I mutation). Asciminib is a first-in-class investigational agent that inhibits BCR-ABL1 kinase activity by Specifically Targeting the ABL Myristoyl Pocket (STAMP). This trial evaluates the safety and efficacy of asciminib to fulfill this unmet clinical need and explore a once-daily (QD) dosing regimen.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Dose-Comparison (Cohort A vs Cohort B) **Blinding:** Open-label **Randomization:** 1:1 for Cohorts A and B (patients without T315I mutation); N/A for Cohort C (patients with T315I mutation)

3.2 Planned Enrollment & Duration Target \$N\$: 115 participants planned (study completed with 56 participants due to enrollment issues) **Number of Sites:** Multicenter (United States) **Estimated**

Study Start: 25-May-2021 **Estimated Primary Completion:** 20-Jun-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients aged ≥ 18 years with a diagnosis of CML in chronic phase (CML-CP). 2. Have shown failure (as per European LeukemiaNet recommendations) or intolerance to the most recent TKI therapy at the time of screening. 3. Prior therapy of ≥ 2 ATP-site TKIs (if T315I mutation is absent) or ≥ 1 prior ATP-site TKI (if T315I mutation is present). 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2.

4.2 Exclusion Criteria 1. Evidence of extramedullary leukemic involvement (with the exception of hepatosplenomegaly). 2. Patients with significant or uncontrolled cardiac disease. 3. Known BCR-ABL1 mutations conferring resistance to the study drug (e.g., V299L).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Cohort A (No T315I): Asciminib 40 mg Twice Daily (BID) - Cohort B (No T315I): Asciminib 80 mg Once Daily (QD) - Cohort C (T315I): Asciminib 200 mg Twice Daily (BID) **Route of Administration:** Oral **Duration of Treatment:** Up to 72 weeks of active treatment

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures to manage co-morbidities as prescribed by the HCP. **Prohibited:** Concurrent administration of other anti-leukemic therapies, unapproved TKIs, or any investigational agents outside the protocol parameters.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -21 to 0	Informed Consent, Medical History, Mutation Analysis, Baseline Labs
Baseline	Day 0	Randomization, First Dose of Asciminib
Interim	Weeks 12, 24, 48	Safety Labs, BCR-ABL1 transcript assessment (RQ-PCR), Efficacy evaluation
End of Study	Week 72	Final Efficacy Assessment (MR4, MR4.5), Exit Interview, IP Accountability
Follow-Up	30 days post-treatment	Safety Follow-Up and final vital signs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The primary endpoint analyses include the incidence and severity of adverse events (AEs), serious AEs, changes in laboratory values, and incidence of notable electrocardiogram (ECG) abnormalities over 24 weeks.

Measurement Method: Clinical laboratory testing, continuous medical monitoring, and MedDRA toxicity scale grading.

7.2 Secondary & Exploratory Endpoints 1. Rate of Major Molecular Response (MMR) ($\leq 0.1\%$ BCR-ABL1/ABL % by international scale) by 12, 24, 48, and 72 weeks. 2. Rate of deep molecular response (MR4 and MR4.5) by 12, 24, 48, and 72 weeks. 3. Progression-free survival (PFS), overall survival (OS), and time to treatment failure.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection evaluating safety profiles, laboratory shifts, and vital sign alterations over 24 weeks and up to the full 72 weeks. **Serious Adverse Events (SAEs):** Evaluated and reported according to standard 24-hour notification protocol for cohort evaluation. **Data Monitoring Committee (DMC):** Overseen by Novartis internal and independent safety monitoring mechanisms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The safety and efficacy analyses included all patients who received ≥ 1 dose of asciminib (Safety Set/ITT equivalent). **Sample Size Justification:** The planned sample size of 115 participants was powered to evaluate safety and determine if the QD (Cohort B) dosing regimen had comparable exposure and safety outcomes to the standard BID (Cohort A) regimen. **Handling of Missing Data:** Standard imputation methods for time-to-event endpoints; failure to record molecular response typically categorized as a non-response for that time interval.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring of safety laboratory variables, RQ-PCR assessments, and vital sign documentation. **Audit Trail:** Utilization of validated electronic Case Report Forms (eCRF) for clinical data lock and querying procedures.

11. Study Identifiers & Governance

Primary ID: CABL001AUS04 **Registry Number:** NCT04666259 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** AIM4CML **Official Regulatory Title:** An Open Label, Multi-center Phase IIb Study of Asciminib (ABL001) Monotherapy in Previously Treated Patients With Chronic Myeloid Leukemia in Chronic Phase (CML-CP) With and Without T315I Mutation

12. Clinical Framework (CML-CP Specific)

Primary Condition: Chronic Myeloid Leukemia in Chronic Phase (CML-CP) **Diagnosis Threshold:** Philadelphia chromosome-positive (Ph+) / BCR-ABL1 positive **Medical Methodology:** Tyrosine Kinase Inhibitor (TKI) / STAMP Inhibitor Therapy **Optimization Target:** Major Molecular Response (MMR) and Deep Molecular Response (DMR)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for CML-CP after TKI failure or intolerance **Education Protocol (HCP):** Protocol training and site initiation visits **Education Protocol (Patient):** Informed consent, mutation analysis explanation, and study drug administration guides **Quality Audit Mechanism:** Clinical site monitoring and ongoing safety data review

14. Observation & Follow-Up Schedule

Total Duration: Up to 72 weeks of active treatment + 30 days safety follow-up **Onsite Visit Cadence:** Baseline, Weeks 12, 24, 48, and 72 **Remote Monitoring Frequency:** As per specific site monitoring plans **Checklist Review Interval:** Routine molecular response (RQ-PCR) assessments at 12-to-24-week intervals

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				